

An extension of the mixed-effects growth model that considers between-person differences in the within-subject variance and the autocorrelation

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Experience sampling methods have led to a significant increase in the availability of intensive longitudinal data. Typically, this type of data is analyzed with a mixed-effects model that allows to examine hypotheses concerning between-person differences in the mean structure by including multiple random effects per individual (eg, random intercept and random slopes). Here, we describe an extension of this model that—in addition to the random effects for the mean structure—also includes a random effect for the within-subject variance and a random effect for the autocorrelation. After the description of the model, we show how its parameters can be efficiently estimated using a marginal maximum likelihood (ML) approach. We then illustrate the model using a real data example. We also present the results of a small simulation study in which we compare the ML approach with a Bayesian estimation approach.

KEYWORDS

experience sampling data, levels, mixed-effects models, within-person autocorrelation, within-person variance

1 | INTRODUCTION

The usage of experience sampling methods leads to an increase in data sets in which many measurements are available for each subject. Typically, this intensive longitudinal data is analyzed with a mixed-effects model¹⁻³ because one can include a number of person-level random effects to model the mean structure of the data. For example, in the linear growth model one can include a random effect for the intercepts and a random effect for the slope of a linear time variable to account for between-person differences in the levels and in the linear rate of change.

A drawback of the mixed-effects model is that all persons are assumed to have the same Level 1 residual variance although research shows that there are systematic differences between persons in the amount they vary around their predicted mean (called within-person variability, see References 4 and 5). For example, in Reference 6, it is found that some individuals are relatively stable with regard to their positive mood, while other individuals are very unstable. Similarly, Geukes et al⁷ found that some people differ in how stable they are in self-esteem, and that the amount of variation depends on participants' values in narcissism. To examine such between-person differences in within-person variability, one can include a random subject effect when modeling the residual variance. Several authors have proposed such extensions of the mixed-effects model, with the mixed-effects location scale model by Hedeker et al^{6,8,9} (MELS) being among the most widely known (but see also References 10 and 11).

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One problem with the previously proposed formulations of the MELS is that—given the random effects—the Level 1 residuals may be autocorrelated over time and are no longer conditionally independent. For experience sampling data, for example, the Level 1 residuals often follow a first-order autoregressive process. The consideration of such a residual covariance structure in the MELS is therefore important for the proper modeling of the data. In fact, one may even consider that persons not only differ in the Level 1 residual variance but also in the autocorrelation. In psychological research, for example, it is assumed that temporal stability actually involves two processes (see Reference 12 for a thorough discussion): within-person variability *and* temporal dependency. While the former refers to the general variability of scores across different time points, the latter describes the general persistence of the scores over time (hence, low scores would reflect instability). To investigate whether these are actually two conceptually different processes or whether they are strongly interdependent, one would have to include a random effect for the autocorrelation in the MELS as well.

We know only a few articles in which such an extension of the MELS has been proposed (see References 13–16). However, all these articles only consider an extension of a random-intercept model, although there may also be time-varying predictors whose influence differ between individuals. Furthermore, a Bayesian approach is used in all cases to estimate the model parameters. The reason for the latter is that a maximum likelihood (ML) estimator for the MELS—at least to our knowledge—has so far only been derived for the case of a mixed-effects model in which the intercept and the weight of a *single* Level 1 predictor is allowed to vary between participants (see Reference 17). Thus, if researchers want to use an ML estimator for the MELS, they then can currently estimate *no* model in which the weights of several predictors vary between individuals, in which the errors are autocorrelated, and/or in which there are also between-person differences in this autocorrelation.

The aim of our article is to remedy this shortcoming by presenting an extension of the MELS that allows for between-person differences in multiple random location effects (ie, a random intercept and multiple random slopes concerning the mean structure), autocorrelated errors following a first-order autoregressive process, and between-person differences in this autocorrelation. Furthermore, we show how the model's parameters can be efficiently estimated using a marginal ML approach. In the following, we first introduce the MELS with the aforementioned extensions. Thereafter, we discuss how the parameters of the model can be estimated using ML. We go on with a real-data illustration of the model. We then present the results of a simulation study in which we compared the performance of the proposed ML approach with a Bayesian approach. Finally, we discuss future research questions.

2 | MIXED-EFFECTS LOCATION SCALE MODEL

In the mixed-effects model, the repeated observations of individual i ($i = 1, \dots, I$ individuals) are written as

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \mathbf{b}_i + \boldsymbol{\epsilon}_i, \quad (1)$$

where the $T_i \times 1$ vector $\mathbf{y}_i$ contains the outcome values of person i of the T_i time points. The matrix $\mathbf{X}_i$ contains the $T_i \times p$ values of the regressors of person i (including a column of 1s for the intercept) and $\boldsymbol{\beta}$ is the corresponding $p \times 1$ vector of regression weights. The predictors in $\mathbf{X}_i$ can be time-constant variables (eg, gender), variables that vary with time (eg, a variable that codes the time point), or interactions of the time-constant and the time-varying variables. $\mathbf{Z}_i$ is a $T_i \times k$ design matrix for the k random effects that are contained in the $k \times 1$ vector $\mathbf{b}_i$. Depending on the type of growth model, $\mathbf{Z}_i$ contains 1s and the values of the time-varying predictors whose coefficients are assumed to vary between individuals. In the linear growth model, for example, $\mathbf{Z}_i$ would contain a column of 1s for the intercept and a variable that codes the measurement time point (eg, 0, 1, $\dots$, $T_i - 1$). $\mathbf{b}_i$ would then contain the corresponding random effect for the intercept and for the slope of the time variable, representing the deviation of individual i from the mean at the first time point (ie, the intercept) and the deviation of i from the average linear growth trend (ie, the slope), respectively. Finally, $\boldsymbol{\epsilon}_i$ is a $T_i \times 1$ vector of residual terms.

Typically, it is assumed the random effects $\mathbf{b}_i$ are normally distributed with expectation zero and covariance matrix $\boldsymbol{\Sigma}_b$. In case of the linear growth model, for example, $\boldsymbol{\Sigma}_b$ is

$$\boldsymbol{\Sigma}_b = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ & \sigma_2^2 \end{pmatrix}, \quad (2)$$

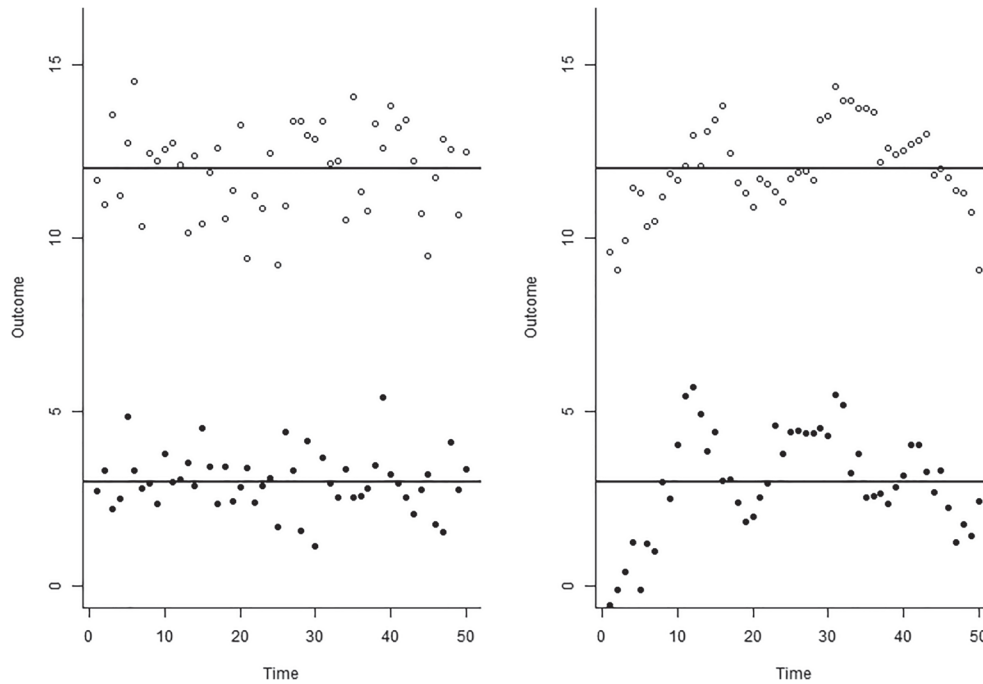


FIGURE 1 Simulated longitudinal data for two individuals with two different autocorrelation parameters (left: $\rho = 0$, right: $\rho = 0.9$) and two different Level 1 residual variance terms (filled circles: $\sigma_{\epsilon}^2 = 1$, empty circles: $\sigma_{\epsilon}^2 = 2$)

where σ_1^2 (σ_2^2) is a measure of the between-person differences in the intercepts (slopes of the linear time variable). σ_{12} describes the degree to which individuals' intercepts and slopes are correlated. A positive covariance, for example, would indicate that individuals with higher intercept terms have larger positive slopes. The error terms are also assumed to be normally distributed with expectation zero and covariance matrix Σ_{ϵ_i} .

In the standard mixed-effects model, the residual covariance matrix is assumed to be equal to $\sigma_{\epsilon_i}^2 \cdot I_{T_i}$ (where I_{T_i} is an $T_i \times T_i$ identity matrix) and it is assumed that the residual variance $\sigma_{\epsilon_i}^2$ is the same for all individuals (ie, $\sigma_{\epsilon_i}^2 = \sigma_{\epsilon}^2$ for all $i = 1, \dots, I$). However, as shown in Figure 1, individuals can differ in the residual variance (eg, for some individuals the variance is larger/smaller than for other individuals). To account for this heterogeneity, one can write the residual variance as a function of person-level covariates and a random-effect for the person:

$$\sigma_{\epsilon_i}^2 = \exp(\mathbf{w}_i^T \boldsymbol{\tau} + \omega_i). \quad (3)$$

Here, $\mathbf{w}_i^T$ is a vector that contains the person-level covariates, $\boldsymbol{\tau}$ is the vector of corresponding weights, and ω_i is a person-specific random effect for the residual variance. It is included so that the residual variance can vary across individuals beyond the effect of the covariates. Most importantly we can allow this random effect to be correlated with the random effects contained in $\mathbf{b}_i$ (see below), reflecting, for instance, that individuals with higher levels tend to have smaller residual variance terms (see References 4 and 5). Finally, the exponential function is used in Equation (4) to ensure that the variance stays positive. When we assume that ω_i is normally distributed, this has also the consequence that the residual variance terms then follow a log-normal distribution (see Reference 6 again).

Equations (1) to (3) correspond with the MELS as suggested in References 6, 8, 9, 18, and 19 and also see Reference 20 and when we assume that there is only one Level 1 predictor whose regression weight can vary between persons (as in the linear growth model example), they would match with the MELS for which an ML estimator was derived in Reference 17. A problem with this formulation of the MELS is that the Level 1 residuals may additionally be autocorrelated over time and this is not considered in the model. For experience sampling data, for example, the Level 1 residuals often follow a first-order autoregressive process. That is, a person's data points do not vary around the person's regression line in a more or less random manner, but rather that a data point that is above (below) the line tend to be followed by a few data points that are also above (below) the line (see the person at the bottom of Figure 1). When we impose this process, the residual covariance matrix (see Reference 2) is

$$\Sigma_{e_i} = \frac{\sigma_{e_i}^2}{1 - \rho_i^2} \begin{pmatrix} 1 & \rho_i & \rho_i^2 & \cdots & \rho_i^{T_i} \\ \rho_i & 1 & \rho_i & \cdots & \rho_i^{T_i-1} \\ \rho_i^2 & \rho_i & 1 & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \rho_i \\ \rho_i^{T_i} & \cdots & \cdots & \rho_i & 1 \end{pmatrix}. \quad (4)$$

Here, $\sigma_{e_i}^2$ is the variance of the Level 1 residual terms and ρ_i is the lag-1 autocorrelation.

As shown in Figure 1, individuals may also differ in the amount their residuals are autocorrelated. To consider this type of heterogeneity, it is possible to write the autocorrelation as a function of person-level covariates.¹¹ Here we go one step further and write the autocorrelation as a function of person-level covariates *and* a random effect for the person (see References 12 and 14):

$$\rho_i = \tanh(\mathbf{r}_i^T \boldsymbol{\gamma} + \kappa_i), \quad (5)$$

where $\mathbf{r}_i^T$ is a vector of person-level covariates, $\boldsymbol{\gamma}$ contains the respective coefficients, and κ_i is person i 's random effect for the autocorrelation. As in the case of the residual variance, this term is included so that the autocorrelation can differ between persons beyond the effect of covariates. It also allows us to model, for example, the covariance between the random effects for the residual variance and the autocorrelation which are assumed to be related (see Reference 12). The tangens hyperbolicus function is used in Equation (5) because we have to ensure that the autocorrelation remains in the interval from -1 to 1 during estimation. By allowing a negative autocorrelation, we follow the literature concerning time series models²¹ or longitudinal mixed-effects models.² Compared to a positive autocorrelation, a negative autocorrelation means that when one observation is above the regression line of an individual, the other observation—depending on the lag—is below the line (and vice versa). Negative autocorrelations occur less often than positive ones, but they can occur and this is the reason why we allow them in the model²¹.

Let $\mathbf{v}_i = (\mathbf{b}_i, \omega_i, \kappa_i)$ a vector that contains all random effects of person i . In the following, we will assume that $\mathbf{v}_i$ is normally distributed with covariance matrix Σ_v . In its most general form Σ_v is

$$\Sigma_v = \left[\begin{array}{c|c} \Sigma_{\mathbf{b}} & \Sigma_{\mathbf{b},(\omega,\kappa)} \\ \hline \Sigma_{\mathbf{b},(\omega,\kappa)} & \Sigma_{(\omega,\kappa)} \end{array} \right], \quad (6)$$

where the covariance block matrix $\Sigma_{\mathbf{b},(\omega,\kappa)}$ contains the covariance terms between the random effects in $\mathbf{b}$ (referring to the mean structure) and the random effect of the residual variance ω or the autocorrelation κ , respectively. $\Sigma_{(\omega,\kappa)}$ contains the variance terms of ω_i and κ_i , respectively, and their covariance. In case of the growth model, to make this more concrete, Σ_v is

$$\Sigma_v = \left[\begin{array}{cc|cc} \sigma_1^2 & \sigma_{12} & \sigma_{1\omega} & \sigma_{1\kappa} \\ & \sigma_2^2 & \sigma_{2\omega} & \sigma_{2\kappa} \\ \hline & & \sigma_{\omega}^2 & \sigma_{\omega\kappa} \\ & & & \sigma_{\kappa}^2 \end{array} \right], \quad (7)$$

where σ_{ω}^2 (σ_{κ}^2) estimates the amount of between-person differences in the residual variance (autocorrelation) and σ_1^2 and σ_2^2 are interpreted as above. The terms $\sigma_{1\omega}$, $\sigma_{1\kappa}$, $\sigma_{2\omega}$, and $\sigma_{2\kappa}$ are covariance parameters describing the relationships between the mean-level related random effects and the scale-related random effects. For example, $\sigma_{1\omega}$ is a measure of the covariance between the random effects concerning the intercepts and the residual Level 1 variance. When this covariance is negative this indicates that higher intercepts go along with smaller residual variance terms. Finally, $\sigma_{\omega\kappa}$ is the covariance of the scale-related random effects. When this covariance is positive this would indicate that higher residual variance terms are associated with higher autocorrelation terms. Empirically, relatively little is known about whether this covariance is zero or not (see the article by Jahng et al¹² again), and hence, whether they reflect different processes. However, the model proposed here allows to examine this covariance in addition to getting estimates of the amount of between-person differences between the scale-random effects.

3 | ESTIMATION

We use a marginal ML approach to estimate the model parameters. To this end, we further assume that the distribution of the observed outcome values of person i , $\mathbf{y}_i$, conditional on the random effects contained in $\mathbf{v}_i$ is also normal with expectation $\boldsymbol{\mu}_{\mathbf{y}_i} = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i$ and covariance matrix $\boldsymbol{\Sigma}_{\epsilon_i}$. The marginal likelihood of the data can be computed by multiplying the likelihood contributions of all participants:

$$L(\boldsymbol{\theta}) = \prod_{i=1}^I \int_{\mathbf{v}_i} f(\mathbf{y}_i | \boldsymbol{\mu}_{\mathbf{y}_i}, \boldsymbol{\Sigma}_{\epsilon_i}, \mathbf{v}_i) f(\mathbf{v}_i | \boldsymbol{\Sigma}_v) d\mathbf{v}_i = \prod_{i=1}^I \int_{\mathbf{v}_i} L_i d\mathbf{v}_i. \quad (8)$$

Here, $\boldsymbol{\theta}$ contains the parameters that we want to estimate. The marginal log-likelihood of the sample is

$$\log L(\boldsymbol{\theta}) = \sum_{i=1}^I \log \left[\int_{\mathbf{v}_i} L_i(\boldsymbol{\theta}) d\mathbf{v}_i \right] = \sum_{i=1}^I \log \left[\int_{\mathbf{v}_i} f(\mathbf{y}_i | \boldsymbol{\mu}_{\mathbf{y}_i}, \boldsymbol{\Sigma}_{\epsilon_i}, \mathbf{v}_i) f(\mathbf{v}_i | \boldsymbol{\Sigma}_v) d\mathbf{v}_i \right]. \quad (9)$$

The standard ML approach consists of solving Equation (9) for the parameter values that maximize the log-likelihood. However, no analytical solution exists for the integrals that appear in the log-likelihood function so that they have to be numerically approximated during estimation. A problem with the approximation is that its computational burden increases very quickly with the number of random effects in $\mathbf{v}_i$. This burden is surely the reason why ML has so far only been suggested for a MELS with a single time-varying variable (or Level 1 predictor).

To decrease the computational burden, we exploit the normality assumption and write the joint density of $\mathbf{v}_i$ as the product of the conditional density $f(\mathbf{b}_i | [\omega_i, \kappa_i], \boldsymbol{\Sigma}_{v_b})$ and $f([\omega_i, \kappa_i] | \boldsymbol{\Sigma}_{(\omega, \kappa)})$. We can then analytically integrate out $\mathbf{b}_i$ so that the marginal distribution of $\mathbf{y}_i$ is

$$f(\mathbf{y}_i) = \int_{\mathbf{t}_i} f(\mathbf{y}_i | \mathbf{t}_i, \boldsymbol{\beta}, \boldsymbol{\tau}, \boldsymbol{\gamma}, \boldsymbol{\Sigma}_b, \boldsymbol{\Sigma}_{b, \mathbf{t}}) \cdot f(\mathbf{t}_i | \boldsymbol{\Sigma}_{\mathbf{t}}) d\mathbf{t}_i, \quad (10)$$

where $\mathbf{t}_i = (\omega_i, \kappa_i)$. $f(\mathbf{y}_i | \mathbf{t}_i, \boldsymbol{\beta}, \boldsymbol{\tau}, \boldsymbol{\gamma}, \boldsymbol{\Sigma}_b, \boldsymbol{\Sigma}_{b, \mathbf{t}})$ is then the density of the (continuous) mixed-effects model. It is thus a multivariate normal distribution whose expectation and covariance matrix is

$$\begin{aligned} \boldsymbol{\mu}_{\mathbf{y}_i} &= \mathbf{X}_i\boldsymbol{\beta} - \mathbf{Z}_i\boldsymbol{\Sigma}_{b, \mathbf{t}}\boldsymbol{\Sigma}_{\mathbf{t}}^{-1}\boldsymbol{\Sigma}_{b, \mathbf{t}}^T(\mathbf{t}_i)^T, \\ \boldsymbol{\Sigma}_{\mathbf{y}_i} &= \mathbf{Z}_i\boldsymbol{\Sigma}_{v_b}\mathbf{Z}_i^T + \boldsymbol{\Sigma}_{\epsilon_i}, \end{aligned} \quad (11)$$

where $\boldsymbol{\Sigma}_{v_b} = \boldsymbol{\Sigma}_b - \boldsymbol{\Sigma}_{b, \mathbf{t}}\boldsymbol{\Sigma}_{\mathbf{t}}^{-1}\boldsymbol{\Sigma}_{b, \mathbf{t}}^T$. The advantage of this reformulation is that it restricts the integral approximation to an integral that is defined across the vector containing the scale-related random effects only, that is, the *two* random effects in $\mathbf{t}$. Thus, the number of random location effects no longer plays a role in the numerical approximation of the integrals.

In our implementation we maximize the marginal likelihood in Equation (8) to obtain the ML estimates of the parameters. We use an gradient-based optimization algorithm (eg, the L-BFGS algorithm in the *nloptr* function in R; see also References 22 and 23) for this, whereby the general form of the gradient is:

$$\frac{\partial \log(L)}{\partial \boldsymbol{\theta}} = \sum_{i=1}^N \frac{1}{f(\mathbf{y}_i)} \int_{\mathbf{v}_i} L_i \frac{\partial \log(L_i)}{\partial \boldsymbol{\theta}} d\mathbf{v}_i. \quad (12)$$

The first derivatives of the log-likelihood for each group of parameters contained in $\boldsymbol{\theta}$ can be requested from the author of the article. Finally, we use adaptive Gaussian quadrature (AGH) to approximate the integrals occurring in the log-likelihood and the gradient (see Reference 24). One can also use other approximation methods (eg, Monte Carlo approximations, see Reference 25, or an H-likelihood approach, see Reference 10), but we found AGH to be very accurate and efficient (see also Reference 26).

4 | ILLUSTRATIVE EXAMPLE

We use data of an experience sampling study to illustrate the suggested models. In this study, $I = 85$ individuals (65 women) completed different personality, affective, and motivational state measures on 82 consecutive days.

TABLE 1 Descriptive information for the variables used in the illustrative example

Variables	<i>M</i>	SD	Min	Max
Age	22.7	5.85	18.0	65.0
No. of measurements	76.9	7.15	33.0	82.0
Neuroticism	4.60	1.31	1.33	7.00
Friendliness (BS)	4.45	0.49	2.29	5.45
Friendliness (WS)	0.00	0.78	−4.05	3.71
Self-esteem	3.79	1.27	1.00	6.00

Note: *I* = 85 individuals (65 women). No. of measurements refer to no. of measurements per participant. *M*, mean; SD, standard deviation; Min, minimum; Max, maximum. BS refers to the between-subject and WS to the within-subject part of a time-varying variable.

TABLE 2 Results concerning the variance components for the illustrative example across four models

Parameters	Standard models			MELS models			E-MELS models	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
σ_1^2	0.47	0.46	0.46	0.47	0.47	0.49	0.46	0.46
σ_2^2	0.04	0.03	0.03	0.04	0.03	0.03	0.03	0.03
σ_ω^2	-	-	-	0.25	0.24	0.24	0.24	0.24
σ_κ^2	-	-	-	-	-	-	0.02	0.02
ρ_{12}	0.33	0.31	0.30	0.33	0.34	0.38	0.32	0.32
$\rho_{1\omega}$	-	-	-	−0.23	−0.21	−0.28	−0.18	−0.18
$\rho_{2\omega}$	-	-	-	0.27	0.32	0.31	0.37	0.41
$\rho_{1\kappa}$	-	-	-	-	-	-	−0.45	−0.45
$\rho_{2\kappa}$	-	-	-	-	-	-	−0.25	−0.28
$\rho_{\omega\kappa}$	-	-	-	-	-	-	0.15	0.15
Deviance	17 072.1	16 847.9	16 812.0	16 525.1	16 333.7	16 331.8	16 280.4	16 278.3

Note: Results are based on an ML approach approximating the integrals with AGH quadrature using 10 quadrature points. To facilitate interpretation, covariance terms are shown as correlations.

Here, we use participants ratings with regard to their self-esteem (ie, “Today, I was satisfied with myself”) and their friendliness (ie, “Today, I was friendly”). Both ratings had to be made on a 6-point Likert-type scale ranging from 1 (*not at all*) to 6 (*very*). Prior to the experience sampling part of the study, all participants filled out different personality measures, intelligence tests, and demographic variables. From these measures, we use participants values in neuroticism and their age in this illustration. Neuroticism was assessed with three items (from the Big Five Inventory SOEP, see Reference 27) that had to be answered on a 7-point scale ranging from 1 (= *does not apply at all*) to 7 (= *applies perfectly*). Finally, we person-mean centered the friendly ratings prior to the analyses. The person means were included in the models as a time-constant predictor. Table 1 presents some descriptive information about the variables.

Eight models were estimated in which subjects nervousness values were regressed on all aforementioned predictors. The first model was a standard mixed-effects model with random effects for the intercept and the slope but no autocorrelation (Model 1 in Tables 2 and 3). The second model included such an autocorrelation (Model 2). The next model was a MELS that additionally included a random effect for the residual variance (but not for the autocorrelation). Again, we fitted a model without (Model 4) and with (Model 5) an autocorrelation. We then fitted the extended MELS (called E-MELS in the tables) that additionally included random effects for both the residual variance and the autocorrelation (Model 7). For all three models, the standard model, the MELS, and the E-MELS, we then added time-constant predictors for the residual variance and the autocorrelation (Model 3, Model 6, and Model 8). All functions used to estimate the model parameters were implemented in R. A working version of the R package can be obtained from the author of this article.

TABLE 3 Results concerning the predictors for the illustrative example across eight models

	Standard models			MELS models			E-MELS models	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
<i>Predicted mean</i>								
Intercept	3.79 (0.08)	3.79 (0.08)	3.79 (0.08)	3.79 (0.08)	3.79 (0.08)	3.79 (0.08)	3.79 (0.08)	3.79 (0.08)
Age	0.12 (0.07)	0.11 (0.07)	0.11 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.07)	0.10 (0.06)	0.10 (0.06)
Neuroticism	−0.19 (0.08)	−0.19 (0.08)	−0.19 (0.08)	−0.19 (0.07)	−0.19 (0.07)	−0.19 (0.08)	−0.19 (0.07)	−0.19 (0.08)
Friendly (BS)	0.38 (0.07)	0.38 (0.07)	0.38 (0.08)	0.37 (0.07)	0.36 (0.07)	0.38 (0.08)	0.38 (0.07)	0.38 (0.08)
Friendly (WS)	0.45 (0.03)	0.44 (0.02)	0.44 (0.02)	0.45 (0.02)	0.43 (0.02)	0.43 (0.02)	0.44 (0.02)	0.43 (0.02)
<i>Residual variance</i>								
Intercept	0.87	0.87	0.72	−0.40 (0.06)	−0.42 (0.06)	−0.43 (0.06)	−0.44 (0.06)	−0.44 (0.06)
Neuroticism	-	-	−0.01	-	-	−0.03 (0.06)	-	−0.03 (0.06)
Friendly (BS)	-	-	−0.01	-	-	−0.03 (0.06)	-	−0.06 (0.06)
<i>Autocorrelation</i>								
Intercept	-	0.19	0.19	-	0.18 (0.01)	0.18 (0.01)	0.18 (0.02)	0.18 (0.02)
Neuroticism	-	-	0.01	-	-	0.01 (0.01)	-	0.01 (0.02)
Friendly (BS)	-	-	0.02	-	-	0.01 (0.01)	-	0.01 (0.02)

Note: Results of the first three models were obtained with the R package `nlme` and the `lme` function using ML estimation. `nlme` does not report standard errors for the variance parameters and the autocorrelation parameters. Results of the remaining models were based on an ML approach approximating the integrals with AGH quadrature using 10 quadrature points. Values in parentheses are standard errors.

Table 2 shows the estimates of the variance components including the deviance of the eight models. We first compared the fit of the models using likelihood-ratio tests. We note that it can be shown that this test is too conservative for testing null hypotheses about variance and covariance terms.^{2,28} However, a simple remedy to this is to divide the P -value obtained from the test by two and we follow this practice here. When we compared the fit of the standard mixed-effects model (ie, Model 1) with the standard MELS (ie, Model 4), we found that the MELS describes the data better than the standard model, $\chi^2 = 547.1$, $df = 3$, $P < .01$. Moreover, adding an autocorrelation lead to a significant improvement in model fit for the standard model (Model 1 vs Model 2), $\chi^2 = 224.2$, $df = 1$, $P < .01$, and the MELS (Model 4 vs Model 5), $\chi^2 = 191.4$, $df = 1$, $P < .01$. Furthermore, although there were only minimal between-person differences with regard to the autocorrelation ($\sigma_k^2 = 0.02$, see Table 2), the E-MELS fitted the data better compared to the MELS (Model 5 vs Model 7), $\chi^2 = 53.3$, $df = 4$, $P < .01$. Finally, adding covariates for the residual variance and for the autocorrelation to the E-MELS lead to no significant improvement in model fit (Model 7 vs Model 8), $\chi^2 = 2.1$, $df = 4$, $P < .36$.

With regard to the fixed effects estimates (see Table 3), we found—when we apply the conventional significance value of .05—that subjects values in neuroticism, their average friendliness, and their daily friendliness ratings were significant predictors of the subjects self-esteem ratings. Specifically, the lower their values in neuroticism and the higher the two friendliness variables, the lower or higher, respectively, their daily self-esteem. The results also showed that the mean variance estimate was $\exp(-0.44) = 0.64$, and the mean autocorrelation parameter was $\text{atanh}(0.18) = 0.19$. Furthermore, and consistent with the results of the likelihood-ratio test, neither neuroticism nor average friendliness had a significant effect on within-person variance terms or within-person autocorrelation terms. Neglecting statistical significance for a moment, the results showed that participants that were more friendly on average had lower within-person variance terms, $\tau_{F(BS)} = -0.06$, $z = -1.13$, and higher autocorrelation parameters, $\gamma_{F(BS)} = 0.01$, $z = 0.57$.*

5 | SMALL SIMULATION STUDY

We performed a small simulation study to evaluate the performance of the ML approach and also to compare it with the performance of a Bayesian approach. The size of the sample was varied by drawing samples of either $I = 75$ or $I = 125$

participants. We also varied the number of time points being either $T_i = 25$ or $T_i = 50$. The true model parameters were set to values that were similar to the real data illustration. Specifically, $\mathbf{X}$ included—in addition to the intercept—a time-constant and time-varying variable. Both variables were simulated from a standard normal distribution. The vector of regression weights was $\boldsymbol{\beta} = (3.50, 0.30, 0.00)$, whereby the weight for the time-constant (time-varying) variable was 0.30 (0.00). For the residual variance and the autocorrelation we also simulated time-constant variables. For the residual variance the weight vector was $\boldsymbol{\tau} = (0.10, 0.00)$ and for the autocorrelation it was $\boldsymbol{\gamma} = (0.20, 0.10)$. We note that the intercept of 0.10 for the residual variance implies an average residual variance of 1.10 and the intercept of 0.20 for the autocorrelation implied an average autocorrelation of about 0.20. Finally, the covariance matrix of the random effects was set to

$$\boldsymbol{\Sigma}_v = \left[\begin{array}{cc|cc} 0.50 & 0.05 & -0.10 & 0.00 \\ & 0.10 & 0.00 & 0.01 \\ \hline & & 0.20 & 0.03 \\ & & & 0.10 \end{array} \right]. \quad (13)$$

The random slope variance refers to the aforementioned time-varying variable whose average weight was 0. We thus used a model in which a Level 1 predictor has no effect on the mean response, but there were between person differences in this coefficient (in contrast to the illustrative data example).

The ML estimator was implemented as described above. To obtain Bayesian estimates, we employed the Mplus software.²⁹ Mplus uses a Gibbs Sampler to approximate the posterior distribution of the model's parameters. We used non-informative priors for all parameters. For the parameters of the mean structure, we used flat Normal distributions, and for the variance and covariance parameters, we specified inverse Gamma or inverse Wishart distributions. For all models, two chains of 10 000 samples were generated from the posterior distributions. The first 5000 samples were discarded for parameter estimation (ie, burn-in period). Prior practice runs showed that this number of iterations and burn-in period resulted in an acceptable effective sample size for all parameters (above 400). We note that the mean and the median resulted in very similar parameter estimates. We therefore decided to use the medians of the posterior distributions as Bayes estimates.

We computed the average parameter estimate, the root-mean square error (RMSE), and the coverage rate to evaluate the ML estimator. The average parameter estimate was computed by averaging the estimates of a parameter in a simulation condition. The RMSE was computed by taking the squared difference of a parameter estimate and the true parameter in each replication of a simulation condition. We then averaged the values for a simulation condition and took the square root of this average. The CR was computed by calculating the 95% confidence interval with the standard error of an estimate in each replication. The coverage was then coded 1 when the true parameter value was included in the interval and 0 when it was not.

Table 4 presents the results of the simulation study. For the ML approach, convergence problems occurred in only 2 replications of the 4000 replications in the four simulation conditions (one in each of the $T = 25$ time point conditions). As can be seen in the table, ML estimates were more accurate, the greater the number of time points. However, average parameter estimates were already very close to the true population parameters when the number of time points was $T = 25$. Furthermore, the RMSE decreased as the number of time points and/or the number of participants increased. The ML approach also yielded acceptable coverage rates for almost all model parameters. The exceptions were the variance parameters, where undercoverage occurred when the number of participants was $I = 75$. Similar results were obtained for the Bayesian approach. The most notable difference between the two approaches occurred for the variance parameters where the Bayes estimates were further away from the true population parameters than the estimates from the ML approach. In addition, the RMSE was slightly larger and the CR was lower. However, the differences between the two approaches became smaller as the number of time points and/or the number of individuals increased.

6 | DISCUSSION

In the present article, we suggested an extension of the mixed-effects location scale model that allows a researcher to include random effects for the means, the within-person residual variance, and the autocorrelation. In addition,

TABLE 4 Results of the simulation study for the ML approach and the Bayesian approach

Est	Par	True	<i>I</i> = 75						<i>I</i> = 125					
			<i>T</i> = 25			<i>T</i> = 50			<i>T</i> = 25			<i>T</i> = 50		
			APE	RMSE	CR	APE	RMSE	CR	APE	RMSE	CR	APE	RMSE	CR
ML	b_{00}	3.50	3.50	0.09	0.94	3.51	0.09	0.94	3.50	0.07	0.94	3.50	0.07	0.94
	b_{01}	0.30	0.30	0.09	0.95	0.30	0.08	0.95	0.29	0.07	0.94	0.30	0.07	0.95
	b_{10}	0.00	0.00	0.04	0.95	0.00	0.04	0.94	0.00	0.03	0.95	0.00	0.03	0.95
	τ_0	0.10	0.10	0.06	0.95	0.10	0.06	0.95	0.10	0.05	0.94	0.10	0.04	0.96
	τ_1	0.00	0.00	0.06	0.94	0.00	0.06	0.94	0.00	0.05	0.94	0.00	0.04	0.94
	γ_0	0.20	0.20	0.05	0.94	0.20	0.04	0.95	0.20	0.04	0.95	0.20	0.03	0.94
	γ_1	0.00	0.00	0.05	0.93	0.00	0.04	0.94	0.00	0.04	0.95	0.00	0.03	0.95
	σ_1^2	0.50	0.49	0.10	0.91	0.49	0.09	0.92	0.49	0.07	0.92	0.49	0.07	0.92
	σ_2^2	0.10	0.10	0.02	0.91	0.10	0.02	0.93	0.10	0.02	0.93	0.10	0.02	0.95
	σ_{12}	0.05	0.05	0.03	0.96	0.05	0.03	0.95	0.05	0.03	0.96	0.05	0.02	0.95
	σ_{ω}^2	0.20	0.19	0.05	0.93	0.20	0.04	0.91	0.20	0.04	0.94	0.20	0.03	0.91
	σ_{κ}^2	0.10	0.11	0.03	0.96	0.11	0.02	0.95	0.11	0.02	0.96	0.11	0.02	0.96
	$\sigma_{\omega\kappa}$	0.03	0.03	0.03	0.94	0.03	0.02	0.96	0.03	0.02	0.94	0.03	0.02	0.96
	$\sigma_{1\omega}$	−0.10	−0.10	0.05	0.95	−0.10	0.05	0.94	−0.10	0.04	0.94	−0.10	0.04	0.95
	$\sigma_{1\kappa}$	0.00	0.00	0.03	0.95	0.00	0.02	0.95	0.00	0.02	0.94	0.00	0.02	0.96
	$\sigma_{2\omega}$	0.00	0.00	0.04	0.96	0.00	0.03	0.95	0.00	0.03	0.94	0.00	0.03	0.95
	$\sigma_{2\kappa}$	0.01	0.01	0.02	0.94	0.01	0.02	0.95	0.01	0.01	0.96	0.01	0.01	0.94
Bayes	b_{00}	3.50	3.50	0.09	0.97	3.51	0.09	0.96	3.50	0.07	0.96	3.50	0.07	0.96
	b_{01}	0.30	0.29	0.09	0.96	0.30	0.08	0.96	0.29	0.07	0.95	0.30	0.07	0.95
	b_{10}	0.00	−0.00	0.04	0.96	−0.00	0.04	0.95	0.00	0.03	0.97	0.00	0.03	0.96
	τ_0	0.10	0.10	0.06	0.97	0.10	0.06	0.97	0.10	0.05	0.95	0.10	0.04	0.96
	τ_1	0.00	0.00	0.06	0.96	−0.00	0.06	0.95	0.00	0.05	0.94	−0.00	0.04	0.95
	γ_0	0.20	0.19	0.05	0.96	0.18	0.04	0.95	0.18	0.04	0.95	0.18	0.04	0.91
	γ_1	0.00	−0.00	0.04	0.94	0.00	0.04	0.96	0.00	0.03	0.96	0.00	0.03	0.96
	σ_1^2	0.50	0.57	0.14	0.91	0.58	0.13	0.89	0.53	0.09	0.96	0.54	0.09	0.94
	σ_2^2	0.10	0.12	0.04	0.89	0.12	0.03	0.91	0.11	0.02	0.92	0.11	0.02	0.92
	σ_{12}	0.05	0.06	0.04	0.96	0.06	0.04	0.95	0.05	0.03	0.97	0.05	0.03	0.95
	σ_{ω}^2	0.20	0.24	0.07	0.90	0.24	0.06	0.90	0.23	0.05	0.90	0.22	0.04	0.92
	σ_{κ}^2	0.10	0.10	0.02	0.98	0.10	0.02	0.98	0.09	0.02	0.97	0.09	0.01	0.96
	$\sigma_{\omega\kappa}$	0.03	0.03	0.03	0.95	0.03	0.02	0.97	0.03	0.02	0.95	0.03	0.02	0.97
	$\sigma_{1\omega}$	−0.10	−0.12	0.06	0.95	−0.12	0.06	0.95	−0.11	0.04	0.96	−0.11	0.04	0.95
	$\sigma_{1\kappa}$	0.00	0.00	0.04	0.97	0.00	0.03	0.97	−0.00	0.03	0.95	−0.00	0.02	0.96
	$\sigma_{2\omega}$	0.00	0.00	0.03	0.96	0.00	0.02	0.96	0.00	0.02	0.95	−0.00	0.02	0.96
	$\sigma_{2\kappa}$	0.01	0.01	0.02	0.95	0.01	0.02	0.95	0.01	0.01	0.96	0.01	0.01	0.95

Note: The column True contains the population parameters. Models were estimated with an ML approach that approximated the integrals with an AGH quadrature with 10 quadrature points or a Bayesian approach. In each of the two 25 time points conditions there was one replication in which the ML estimator did not converge.

Abbreviations: Est, estimator; Par, parameter.

covariates are allowed to influence all three types of random effects. We also explained how an ML approach can be used to estimate the model's parameters and we presented an empirical example that illustrated the suggested approach. Finally, we reported the results of a small simulation study that investigated the statistical properties of the suggested ML estimator.

From our point of view, the extended model can be used to investigate a number of interesting research questions beyond the ones described in the illustrative example. For instance, and as mentioned in the introduction, an important question with regard to a person's stability is whether it involves a process referring to variability and/or a process referring to temporal dependency. The model proposed here allows to consider both processes separately, to determine their dependency (via the covariance of the respective random effects), and also whether they are influenced differently by different covariates. This allows stability to be better understood both empirically and conceptually.

We used an ML approach to estimate the model's parameters. This contrasts the present article from other articles in which a Bayesian approach was used for related models.^{13-15,23,30-34} Although the ML approach has advantages in comparison to a Bayesian approach, it also has disadvantages such as its computational practicality. For example, we used an AGH procedure here to numerically approximate the integrals. This approximation is better, the higher the number of quadrature points. However, the larger the number points, the larger is the computation time. Also, although not an issue here, convergence problems can occur, for example, when the number of measurements is small. For these reasons, we believe that future research should investigate whether alternative approaches, such as a Monte Carlo EM algorithm,²⁵ variational approximation³⁵ or an H-likelihood approach,¹⁰ are able to decrease the computation time and whether they can solve convergence issues of the ML estimator.

Concerning model development, we think that an interesting avenue for future research would be to extend the model so that the estimation of the autocorrelation does not rest on the assumption of equally spaced time intervals. Although, this assumption is justified in our real data example, there are also studies in which the time intervals are not equally spaced. Also considering a time variable that reflects the person-specific time intervals, does not seem too costly from an algorithmic perspective. Furthermore, future research should examine how the model would behave in case of missing data. In our model extension, we focused on a first-order autoregressive process. We think that considering alternative autocorrelated error structures—such as moving average (MA) or autoregressive moving average (ARMA)—is also a topic that should be considered in future research. As in the case of the AR structure, one could extend the MELS to include these structures, or one could develop an extended MELS in which the parameters of these error structures—in addition to the Level 1 residual variance—are allowed to vary between individuals. Finally, we assumed here that the outcome variable is continuous. Another interesting extension might then be to extend the models to ordinal outcomes.⁸ However, such an extension requires a more complicated estimation algorithm (because of the nonlinear link function), but the model would be more flexible and would allow to investigate some interesting applied and methodological research questions (eg, how much spread across categories is needed for the reliable estimation of random scale effects).

To summarize, the present article proposed an extension of the mixed-effects location scale model and derived an ML estimator that allows multiple random-locations effects (in the MELS or the extension of the MELS). We believe that the described model is of interest to applied researchers and that it raises a number of interesting research questions, the solutions of which promise further interesting model developments.

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DATA AVAILABILITY STATEMENT

The data are made available in an OSF project: <https://osf.io/4zk9y/>.

ENDNOTE

*We also fitted a model to the data in which the covariance between the two random scale effects (ie, σ_{ω_K}) was set to zero. However, the results of this model were largely identical to the results in which the covariance was not constrained to be zero.

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